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Supplementary Materials for

A Soft Wearable System for Real-time Respiratory Disease Diagnosis and Vibrotactile Intervention

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Other Supplementary Materials for this manuscript include the following:

Movies S1 to S5

Supplementary Note 1. Magnetic-field distribution, force generation, and actuation efficiency of the coil-magnet module. The actuation force generated and its efficiency are governed by the geometry of the magnet-coil pair, as well as the magnetic and electrical properties of the module. With the given permanent magnet dimensions (diameter 3 mm, thickness 1 mm), the actuation force and its conversion efficiency can be theoretically modeled by deriving the axial magnetic field produced by the coil, the resulting force exerted on the axially magnetized magnet, and the electrical energy given during actuation.

Given a cylindrical coil with inner radius R_i, outer radius R_o, axial length L_c, total number of turns N (**Supplementary Fig. 12**), and driving current I, the coil occupies the volume $V_{coil} = \pi(R_0^2 - R_i^2)L_c$. Assuming uniform volumetric turn density,

$$v_0 = \frac{N}{V_{coil}}$$

and based on the Biot-Savart law, the axial magnetic field produced by a differential loop of radius r at axial offset $(z - z')$ is

$$B_z^{loop}(z; r, z') = \frac{\mu_0 I r^2}{2[r^2 + (z - z')^2]^{3/2}} dr dz'$$

Integrating over the entire winding volume yields the total axial field:

$$B_z(z) = \mu_0 I v_0 \int_{-L_c/2}^{L_c/2} \int_{R_i}^{R_o} \frac{r^2}{2[r^2 + (z - z')^2]^{3/2}} dr dz'$$

For compactness, the integral can be expressed using a dimensionless geometry factor G

$$B_z(z) = \frac{\mu_0 I N}{2L_c(R_o - R_i)} G(z; R_i, R_o, L_c)$$

where G depends solely on coil dimensions and axial position.

A cylindrical permanent magnet of radius r_m, thickness L_m, and remanence B_r is placed along the coil axis. Then, its magnetic dipole moment is

$$m = MV_m \approx \frac{B_r}{\mu_0} \pi r_m^2 L_m$$

where M is the magnetization. Then, force derivation starts with magnetic potential energy,

$$U = -\int_{V_m} \mathbf{M} \cdot \mathbf{B}\, dV$$

Assuming that $B_z(z)$ varies slowly over the thickness L_m, a Taylor expansion around the magnet center z_c gives

$$U(z_c) \approx -mB_z(z_c) + \mathcal{O}(L_m^3 B_z''')$$

The vertical force follows from

$$F_z = -\frac{\partial U}{\partial z} \approx m\frac{dB_z}{dz}|_{z_c}$$

Alternatively, integrating the field difference across the magnet faces yields

$$F_z = \left(\frac{B_r}{\mu_0}\right)\pi r_m^2[B_z(z_2) - B_z(z_1)]$$

where z_1 and $z_2 = z_1 + L_m$ denote the lower and upper faces of the magnet. With the $B_z(z)$ derived above, the force generated can be expressed as

$$F_z = \frac{\pi r_m^2}{2L_c(R_o - R_i)}[G(z_2) - G(z_1)] \times B_r \times (NI)$$

This expression reveals the contribution of both material properties (B_r), magnet geometry (r_m, L_m), coil configuration (N, R_i, R_o, L_c) and the drive current I.

By defining a force-based efficiency metric as

$$\eta_F = \frac{F_z}{P}$$

where F_z is the normal actuation force on the magnet and $P = I^2 R_{coil}$ is the electrical power dissipated in the coil at current I.

Supplementary Note 2. Actuation pattern control using designated waveform modes. The vibrotactile actuation is generated by driving the coil-magnet module with a carrier oscillation whose amplitude and frequency are modulated according to the designated wave mode. All patterns are produced as a combination of a carrier sinusoidal vibration and time-varying envelope

$$x(t) = A(t)\sin(2\pi f_{carrier}(t)t)$$

where $A(t)$ denotes the amplitude envelope, and $f_{carrier}(t)$ denotes the carrier frequency or its time-varying equivalent. This formulation enables a unified framework for generating diverse haptic cues. The system supports various kinds of wave modes such as spike, envelope, superposition, and melody, each realized by defining a distinct modulation function $A(t)$ or frequency profile $f_{carrier}(t)$.

1. Spike mode (pulsed actuation)
Spike mode delivers a train of short bursts produced by gating the carrier sinusoid with a binary (0/1) envelope. For a spike duration τ_{on}, off-interval τ_{off}, the total number of spikes N

$$A_{spike}(t) = \begin{cases} 1, & 0 \le t - kT_{spike} \le \tau_{on}, \\ 0, & otherwise \end{cases} \qquad k = 0, \dots, N-1$$

with $T_{spike} = \tau_{on} + \tau_{off}$. Then, the generation vibration is

$$x_{spike}(t) = A_{spike}(t)\sin(2\pi f_0 t)$$

where f_0 is the fixed carrier frequency, $f_0 \in (50, 150)$.

2. Envelope mode (amplitude-modulated waveform)
Envelope mode produces smooth rhythmic cues by modulating the carrier with a slowly varying envelope that contains both linear ramping and harmonic modulation. The implemented form is

$$A_{env}(t) = \alpha(\frac{t}{T})[1 - \cos(2\pi f_0 t)]$$

where T is the total envelope duration, f_0 is the fixed carrier frequency, and α determines the maximum duty. The linear term t/T creates a gradual change in intensity, while the cosine term generates an envelope of rise-fall pattern.

$$x_{env}(t) = A_{env}(t)\sin(2\pi f_0 t)$$

3. Fourier superposition mode (user-specific breathing reconstruction)
This mode reconstructs the user's breathing profile by representing the inhalation and exhalation envelopes as a Fourier-based multicomponent superposition of harmonic waveforms. The coefficients are obtained through calibrated measurement and fitting (**Supplementary Note 3**), while the resulting waveform reproduces the smooth temporal evolution of respiratory phase.

For inhalation,

$$A_{inh}(t) = \theta_0^{(in)} + \sum_{k=1}^{K} \left[\theta_{2k-1}^{(in)} \cos\left(\frac{2\pi kt}{T_{inh}}\right) + \theta_{2k}^{(in)} \sin\left(\frac{2\pi kt}{T_{inh}}\right) \right]$$

And for exhalation,

$$A_{exh}(t) = \phi_0^{(ex)} + \sum_{k=1}^{K} \left[\phi_{2k-1}^{(ex)} \cos\left(\frac{2\pi kt}{T_{exh}}\right) + \phi_{2k}^{(ex)} \sin\left(\frac{2\pi kt}{T_{exh}}\right) \right]$$

Therefore, the final actuation pattern is

$$x_{sup}(t) = A_{sup}(t) \sin(2\pi f_0 t)$$

4. Melody mode (frequency-modulated actuation, square waveform)
Melody mode varies tactile percepts by changing the carrier frequency according to the note of melody. For the n-th note of the frequency f_n and duration d_n

$$x_{mel}(t) = A_0 \sin\big(2\pi f_n (t - t_n)\big), \quad t_n \leq t < t_n + d_n$$

Supplementary Note 3. Breathing profile fitting using a Fourier-based multicomponent superposition mode. To reconstruct a smooth and continuous representation of the respiratory envelope, the inhalation and exhalation segments of the breathing cycle were individually fitted using a Fourier-based multicomponent superposition model. This approach is well-suited for physiological waveforms, which are inherently composed of multiple harmonic components rather than a single dominant mode. The resulting fitted curves provide a compact representation of each phase and allow the firmware to reproduce personalized vibrotactile breathing patterns using a small number of coefficients.

Given a segmented inhalation signal $y_{inh}(t)$ sampled over a time interval $[t_0, t_{end}]$, the time axis was first normalized to a unit interval

$$\tau = \frac{t - t_0}{t_{end} - t_0}, \quad 0 \leq \tau \leq 1$$

The amplitude was also normalized to unify for numerical stability.

$$\bar{y}_{inh}(\tau) = \frac{y_{inh}(\tau)}{\max\,(y_{inh})}$$

The same normalizeation was applied to the exhalation signal $y_{exh}(t)$ to obtain $\bar{y}_{exh}(\tau) \in (0,\ 1)$. Then, each phase (inhale or exhale) was modeled as a finite Fourier series

$$\hat{y}(\tau) = \theta_0 + \sum_{k=1}^{K} [\theta_{2k-1} \cos(2\pi k\tau) + \theta_{2k} \sin(2\pi k\tau)\,]$$

where K is number of harmonics (in this work 6), and $\theta = \{\theta_0, \theta_1, \dots, \theta_{2K}\}$ is the coefficient vector.
To identify the coefficients, a design matrix Φ was constructed as,

$$\Phi = \begin{bmatrix} 1 & \cos(2\pi\tau_1) & \sin(2\pi\tau_1) & \cdots & \cos(2\pi K\tau_1) & \sin(2\pi K\tau_1) \\ 1 & \cos(2\pi\tau_2) & \sin(2\pi\tau_2) & & \cos(2\pi K\tau_2) & \sin(2\pi K\tau_2) \\ & \vdots & & \ddots & & \vdots \\ 1 & \cos(2\pi\tau_N) & \sin(2\pi\tau_N) & \cdots & \cos(2\pi K\tau_N) & \sin(2\pi K\tau_N) \end{bmatrix}$$

Where N is the number of data samples in the inhalation (or exhalation) segment and the coefficient vector was obtained via least-square regression

$$\theta = arg\,\min_{\theta} \|\Phi\theta - \bar{y}\,\|_2^2 = (\Phi^T\Phi)^{-1}\Phi^T\bar{y}$$

This procedure was performed separately for the inhalation and exhalation segments, yielding two coefficient sets $\theta_{inh}, \theta_{exh}$.
The reconstructed envelopes were then computed as

$$\hat{y}_{inh}(\tau) = \Phi_{inh}\theta_{inh} \quad and \quad \hat{y}_{exh}(\tau) = \Phi_{exh}\theta_{exh}$$

Supplementary Fig. 17 shows representative examples of the measured and fitted curves, illustrating the accuracy of the harmonic superposition model in capturing the underlying respiratory dynamics.

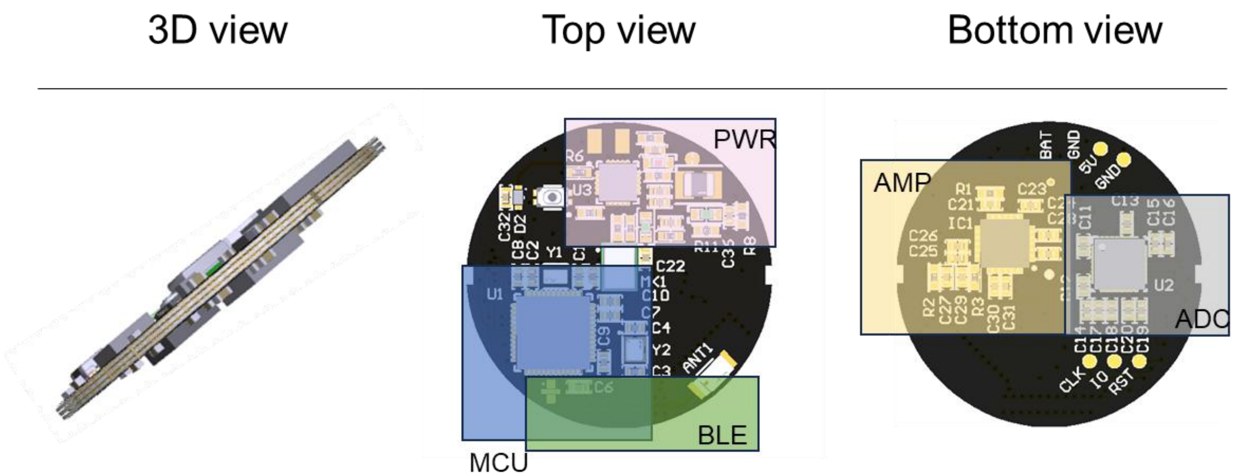


Fig. S1. Printed circuit board layout with footprints

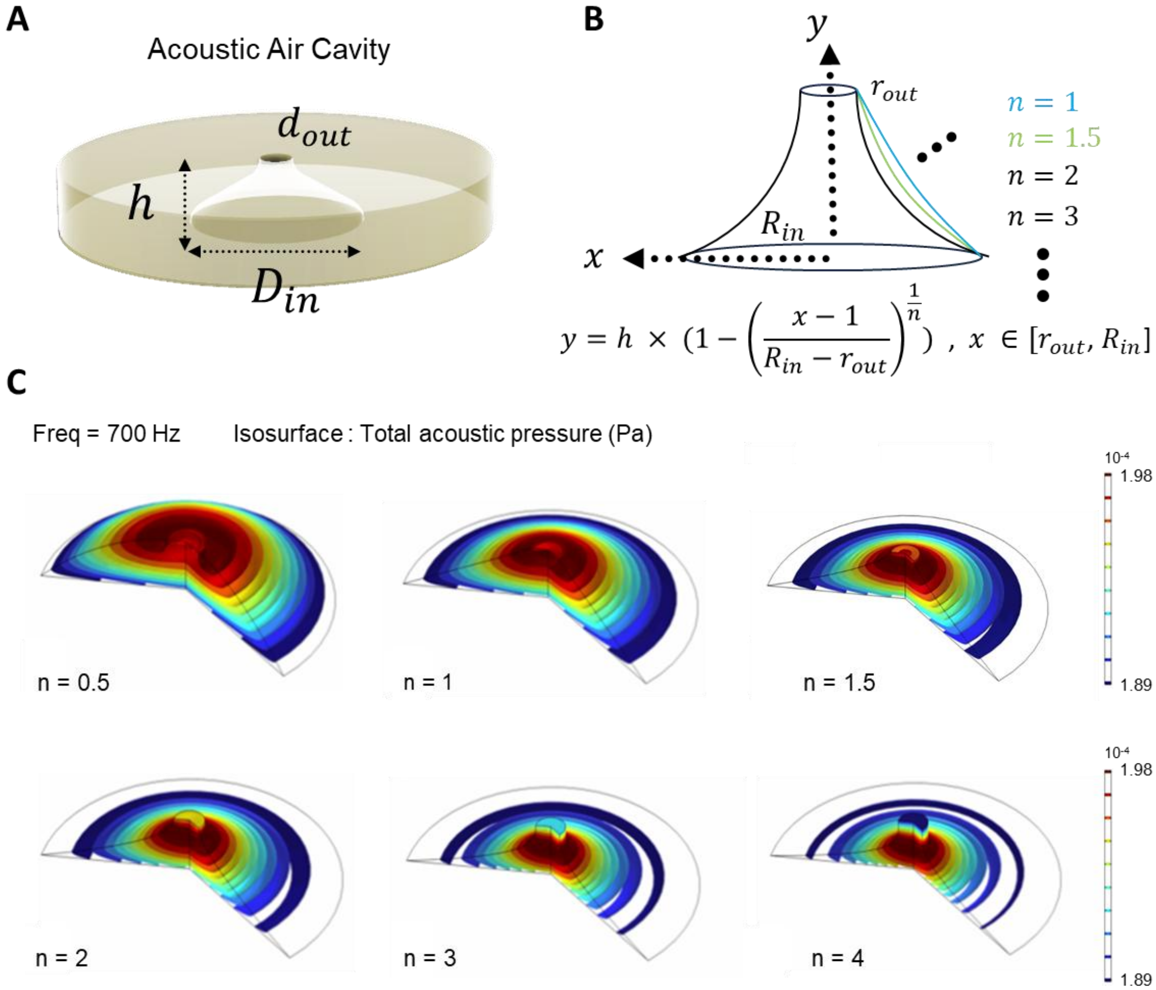


Fig. S2. Acoustic air cavity geometry and curvature-dependent pressure field distribution. (A) Design parameters of the acoustic air cavity, **(B)** Internal wall curvature profile, **(C)** Acoustic pressure distribution across curvature gradient.

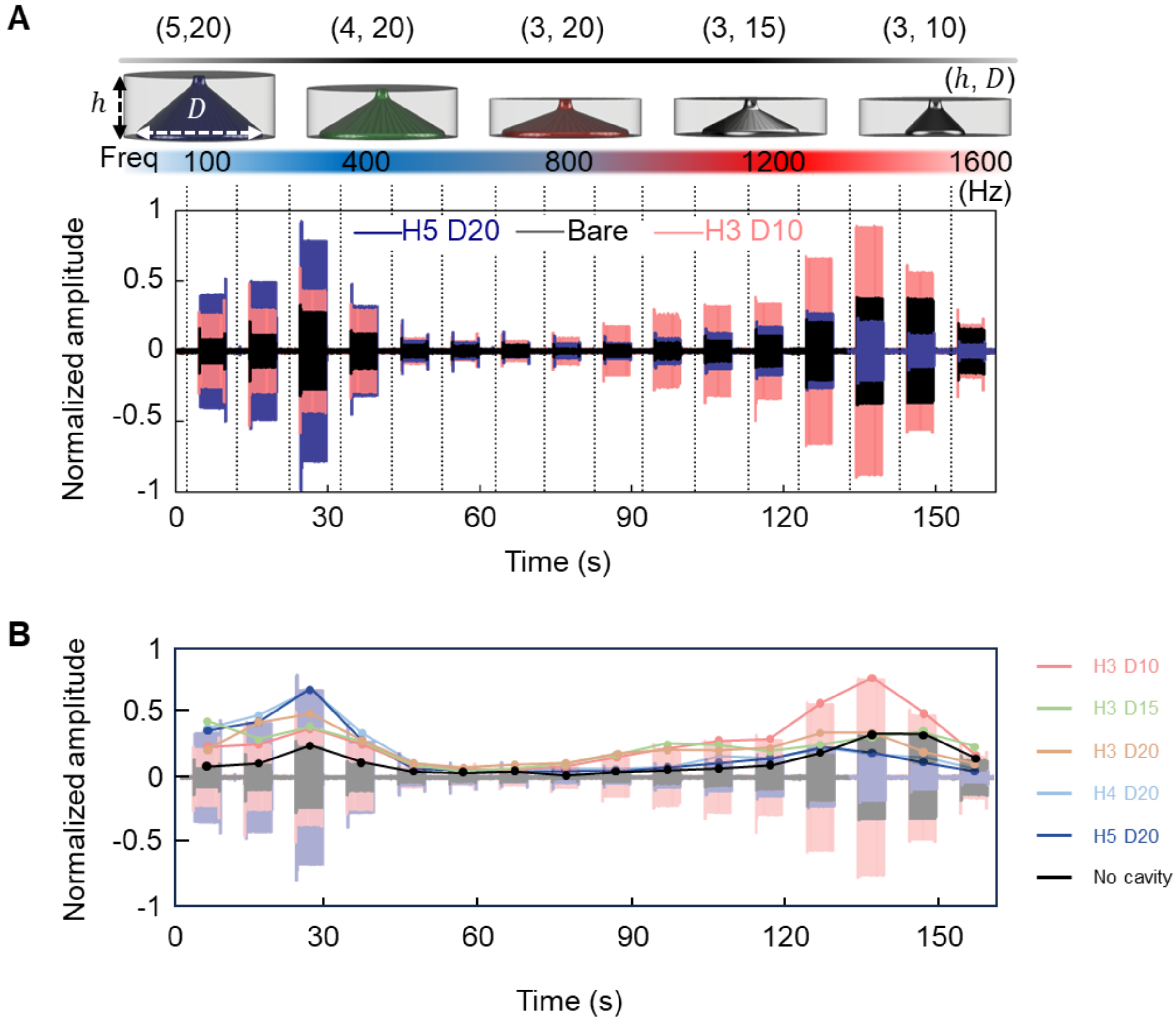


Fig. S3. Frequency response plots of the acoustic cavity structure. (A) Frequency response measurements with different acoustic cavity designs. **(B)** Comparing 6 configurations: no cavity, H5 D20, H4 D20, H3 D20, H3 D15, and H3 D10.

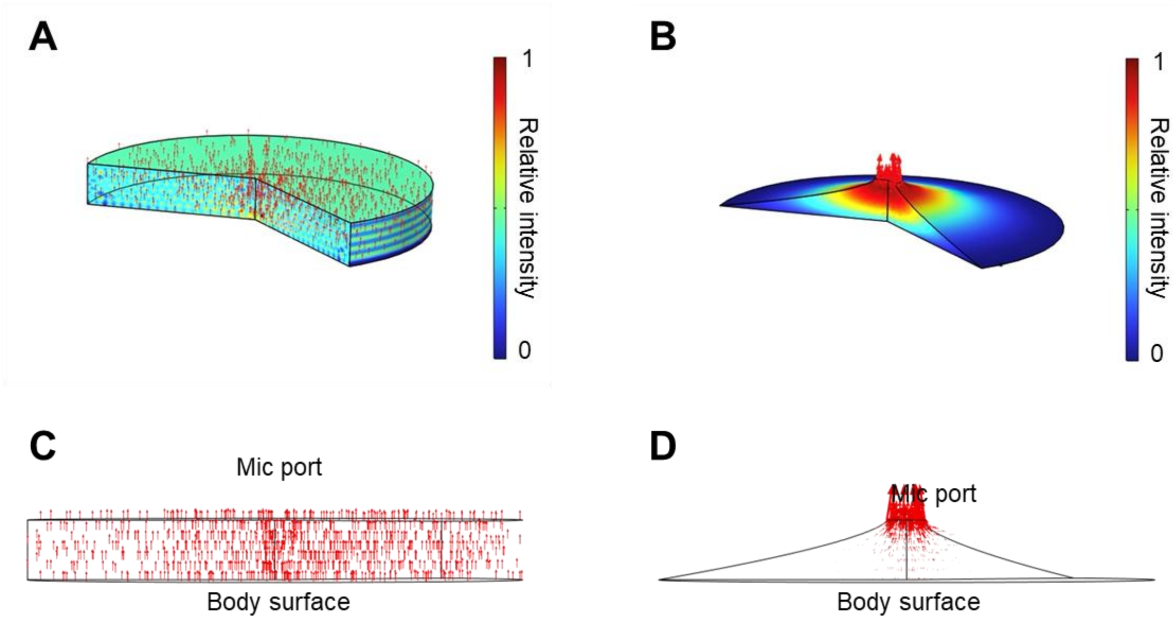


Fig. S4. Sound pressure level and airflow distribution in tubular and cavity structures. Sound pressure level (dB) distribution within (**A**) the tubular structure and (**B**) the air cavity. Airflow volume distribution across the (**C**) tubular structure and (**D**) the air cavity from the body surface toward the microphone port.

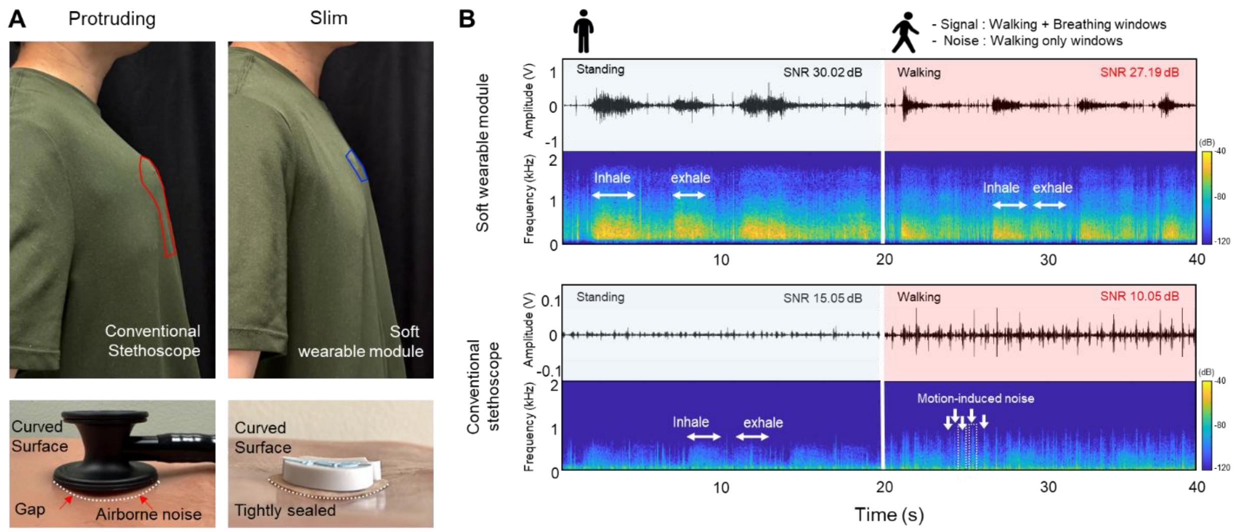


Fig. S5. Comparison between a conventional stethoscope and our soft wearable module. (A) Geometric conformity and slim on-body interface **(B)** Representative lung-sound waveforms and time-frequency spectrogram recorded during standing and walking conditions.

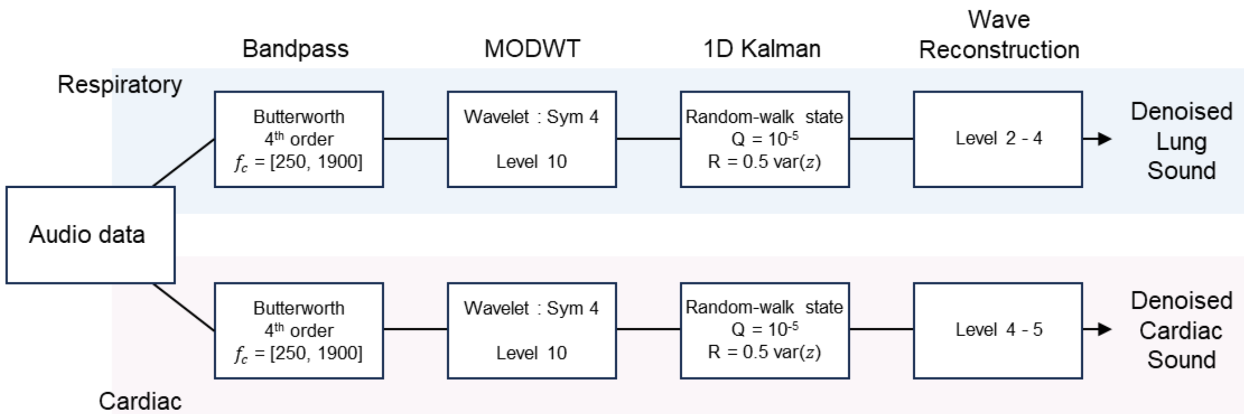


Fig. S6. Signal processing pipeline for denoising acoustic measurements.

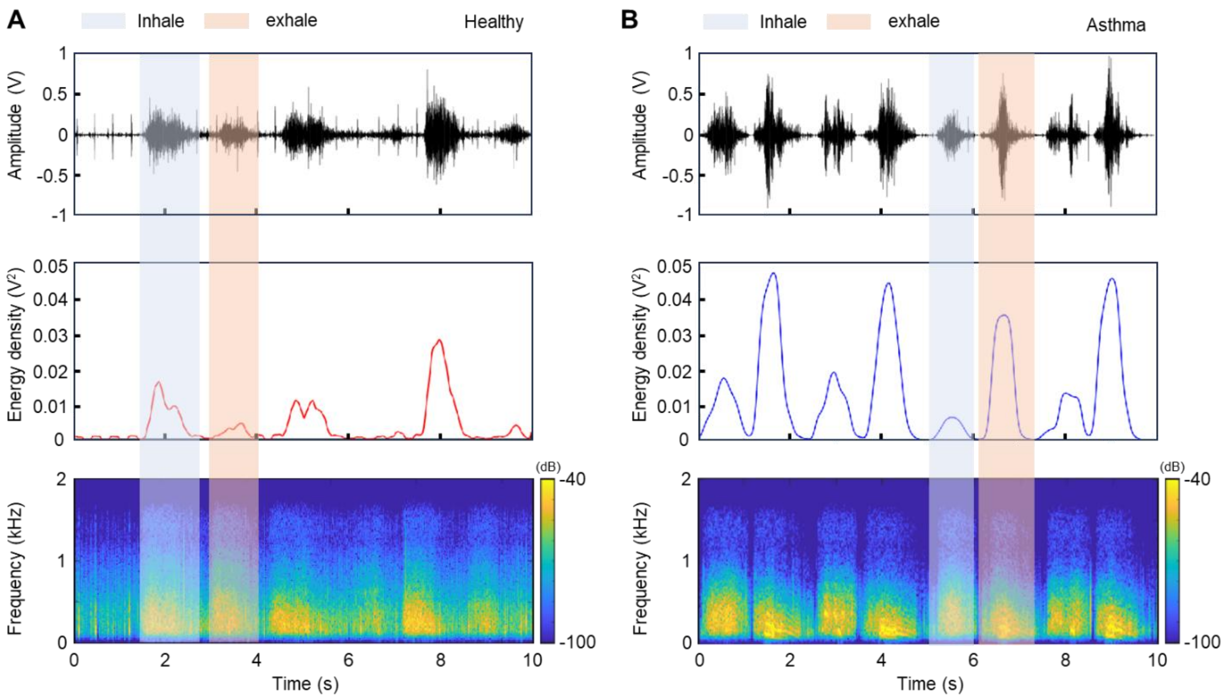


Fig. S7. Comparison of breathing sound measured between healthy and asthma cases.

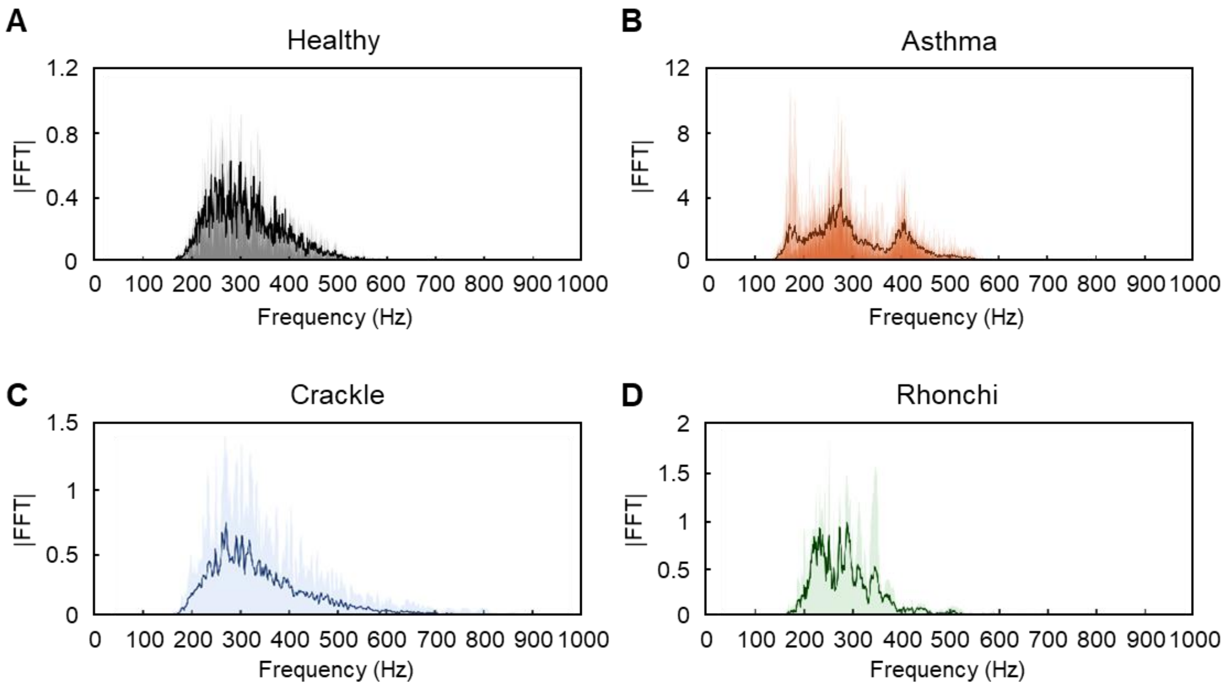


Fig. S8. Representative frequency spectra of respiratory sounds under different conditions. (**A**) Healthy subject: dominant frequency components concentrated between 200–500 Hz with relatively low amplitude. (**B**) Asthma: presence of multiple peaks with higher overall power, particularly near 280 Hz and 400 Hz, indicative of wheezing events. (**C**) Crackle: broad frequency distribution with pronounced fine/coarse crackles. (**D**) Rhonchi: lower-frequency, narrow-band energy (<300 Hz) consistent with sonorous wheezes. Shaded regions represent overlaid FFTs of multiple breathing cycles, and bold lines indicate the mean spectra for each condition.

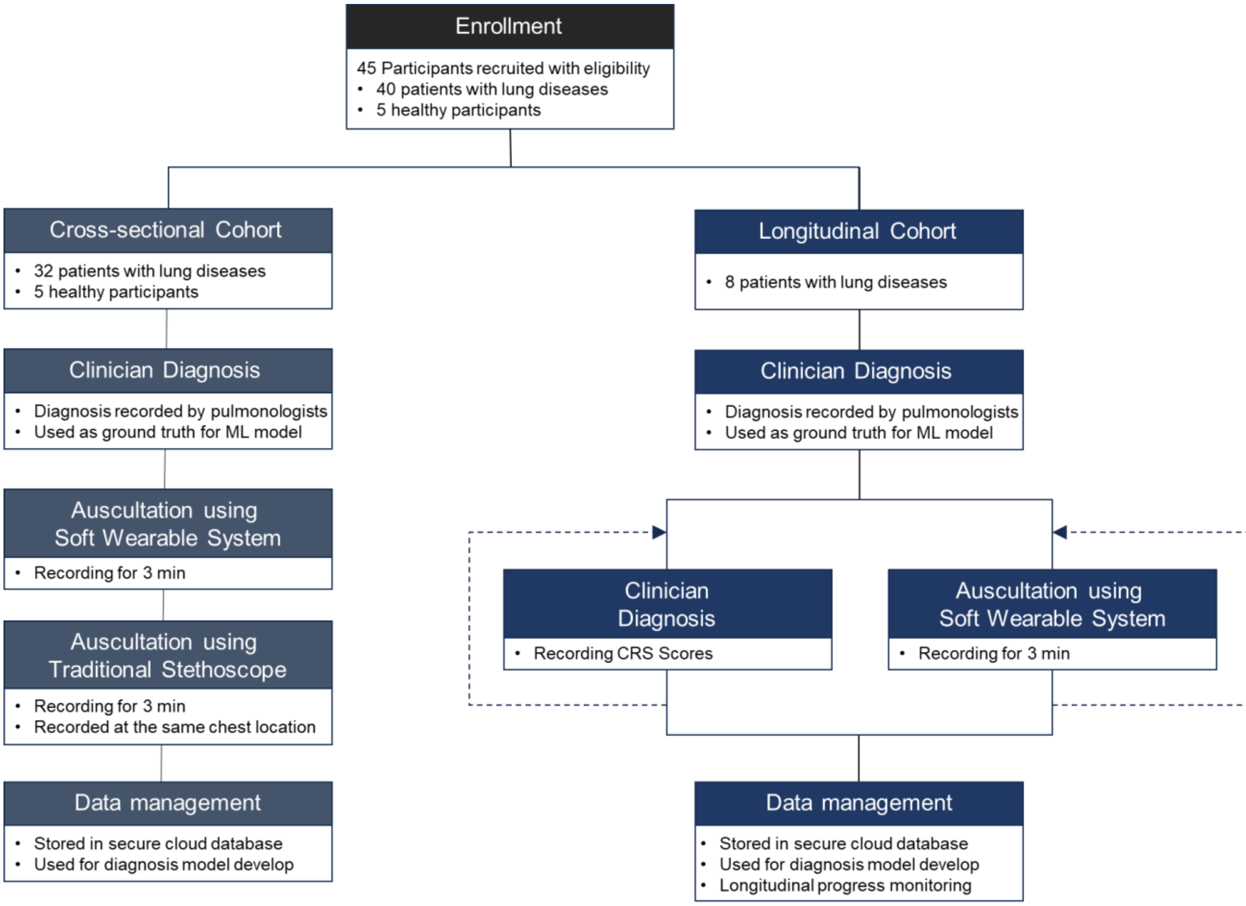


Fig. S9. Overview of participant enrollment, cohort stratification, and clinical study protocol for cross-sectional and longitudinal cohorts.

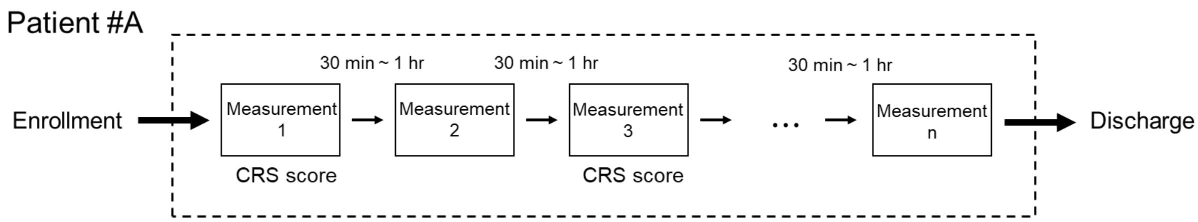


Fig. S10. Study protocol for longitudinal measurement for progress monitoring. Critical Respiratory Score (CRS) represents clinician's evaluation of the patient's respiratory status.

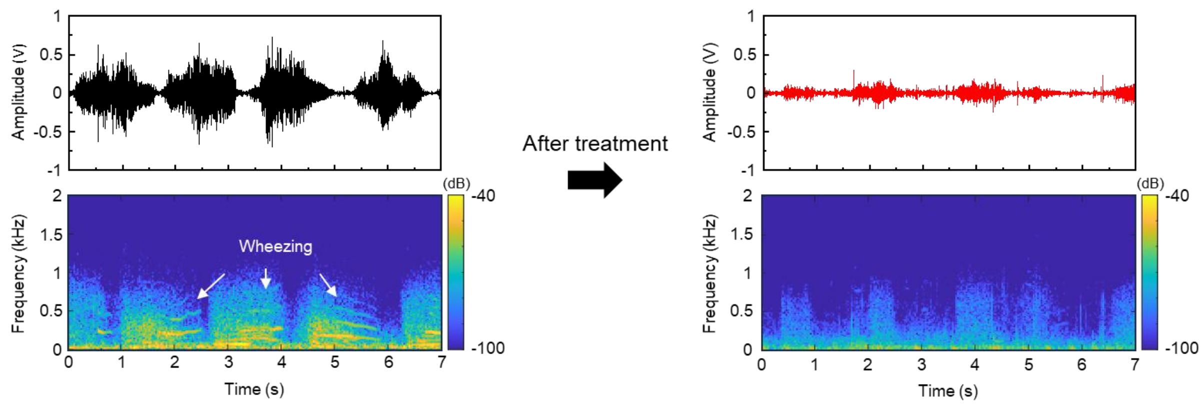


Fig. S11. Comparison of breathing sound measured between before treatment and after treatment

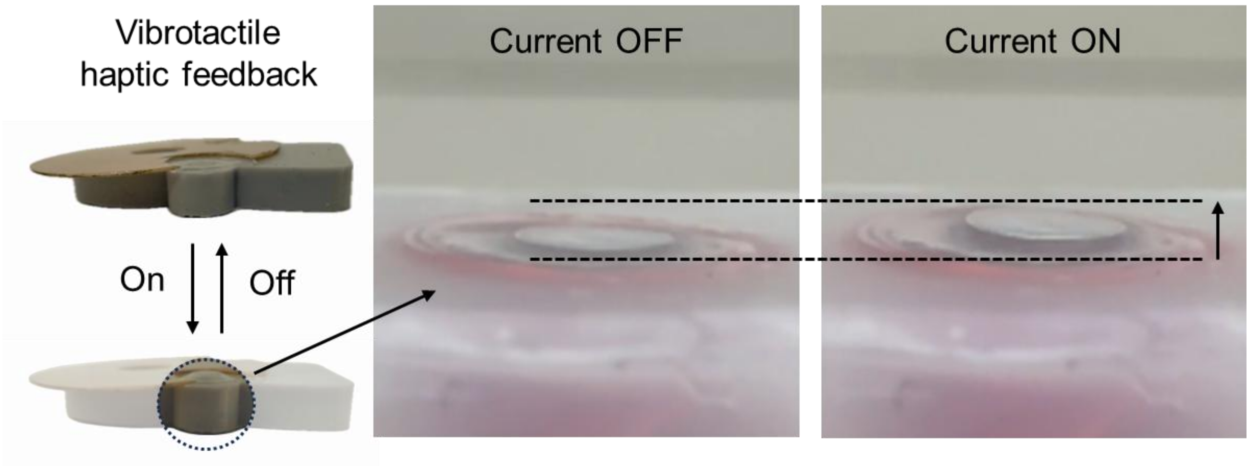


Fig. S12. Reciprocating vertical motion generated during vibrotactile feedback.

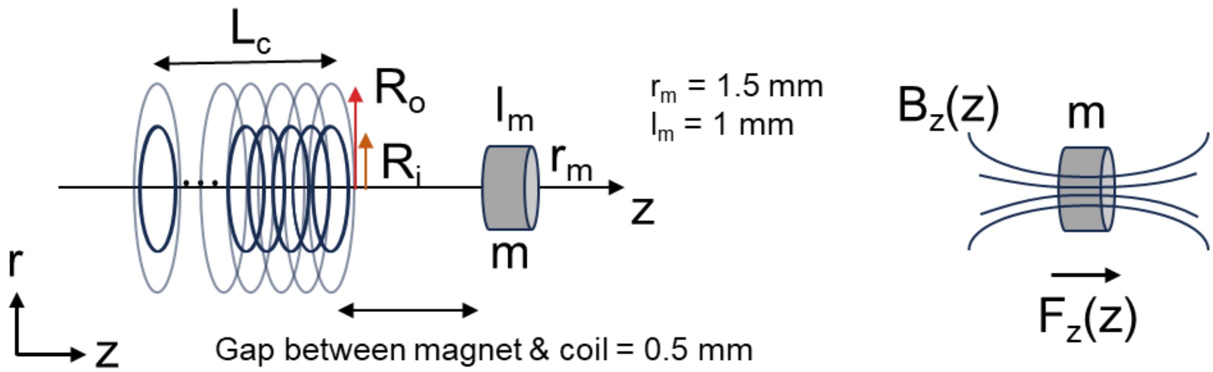
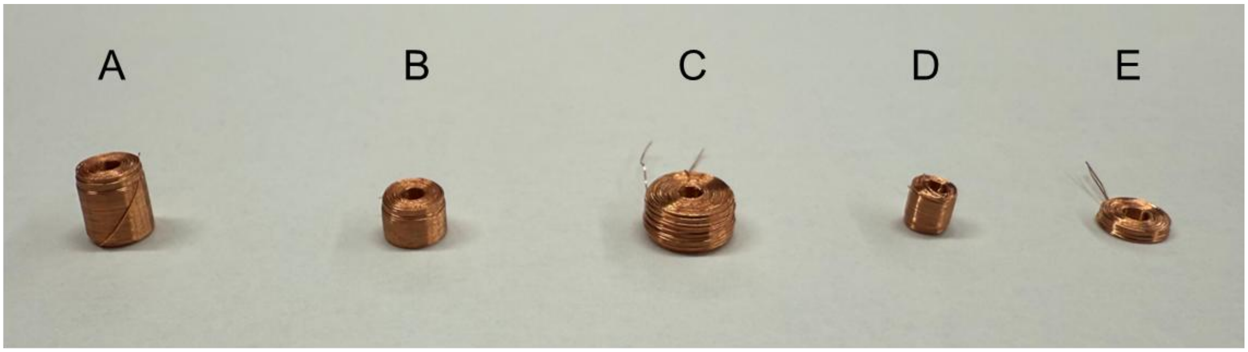


Fig. S13. Electromagnetic actuation mechanism of the coil-magnet module.



Coils	Outer diameter (mm)	Inner diameter (mm)	Length (mm)	Resistance (Ω)
A	4.2	1.5	5.0	28.4
B	4.2	1.5	3.0	17
C	6	1.5	3.0	35
D	3	1.5	3.0	8.3
E	4.2	1.5	1.0	5

Fig. S14. Fabricated coils and design parameters.

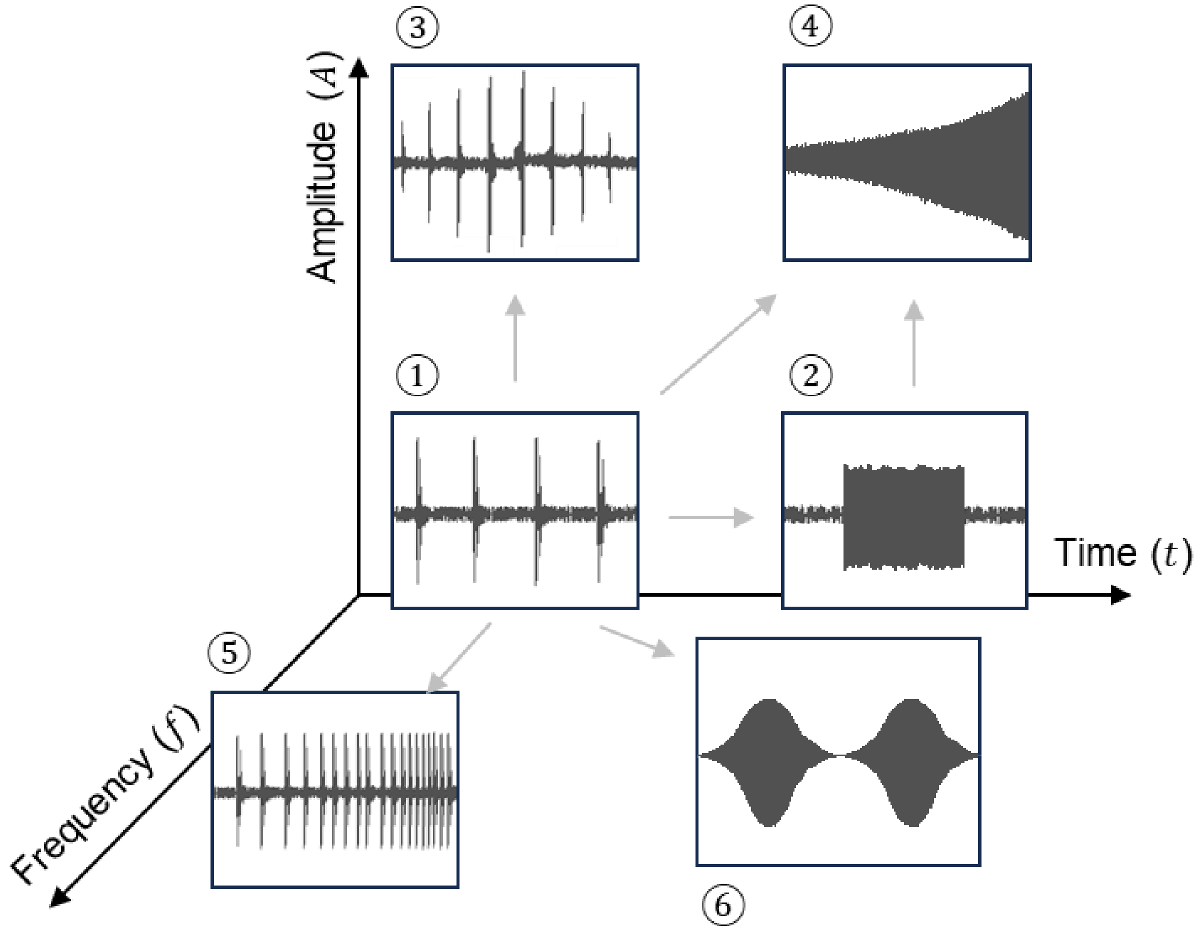


Fig. S15. Parameter-dependent generation of vibrotactile modes. This includes high-frequency pulse mode, spike mode, amplitude-modulated mode, broadband burst mode, regression mode.

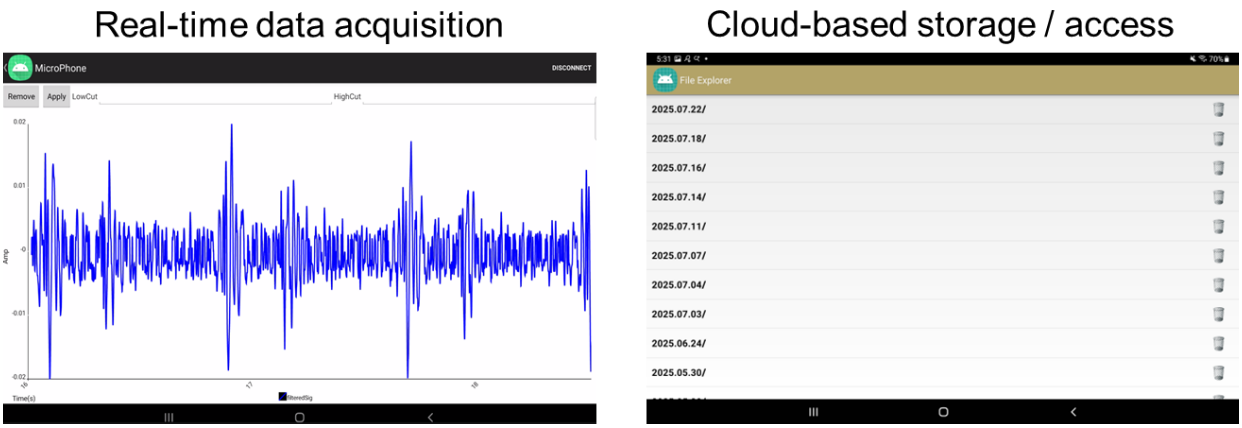
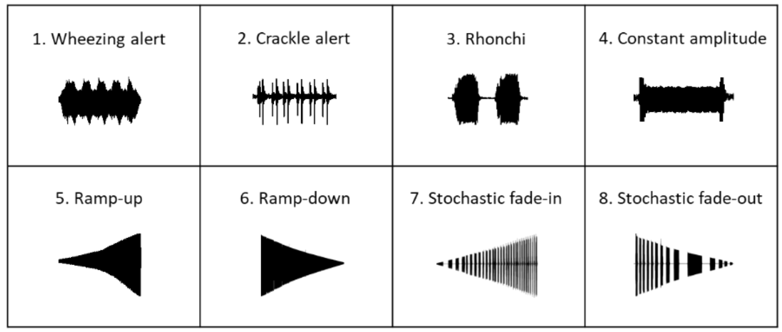


Fig. S16. App used for real-time wireless data acquisition and cloud-based data storage and access



	1	2	3	4	5	6	7	8
1	58	0	0	2	0	0	0	0
2	0	60	0	0	0	0	0	0
3	0	0	60	0	0	0	0	0
4	0	0	0	52	5	3	0	0
5	0	0	0	0	60	0	0	0
6	0	0	0	7	0	53	0	0
7	0	0	0	0	0	5	51	4
8	0	0	0	0	2	0	0	58

Fig. S17. Confusion matrix showing the recognition accuracy of eight vibrotactile actuation patterns evaluated in healthy participants (n = 5).

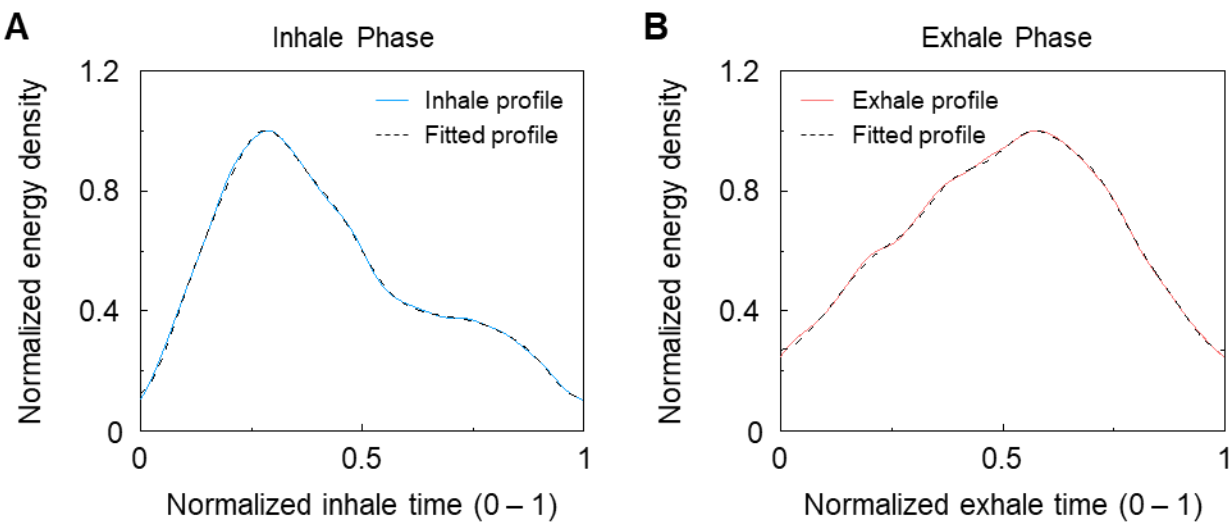


Fig. S18. Representative amplitude profiles that are fitted from measurements for of personalized guided-breathing patterns.

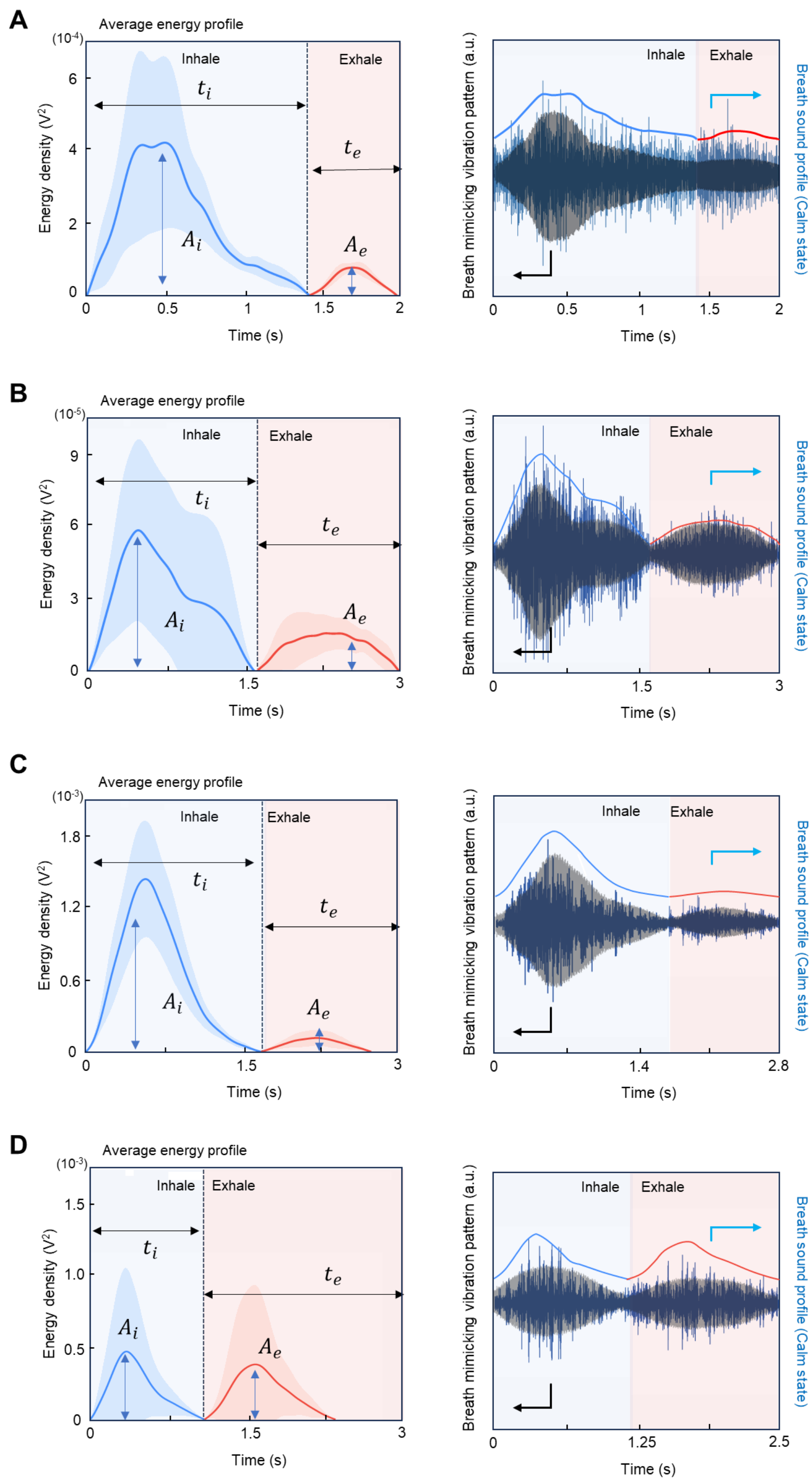


Fig. S19. Breathing-pattern extraction and tactile reconstruction across healthy participants. Left part of each figure shows the extraction of inhalation–exhalation breathing envelopes from respiratory sounds and the right part of each figure shows corresponding reconstruction of personalized vibrotactile patterns.

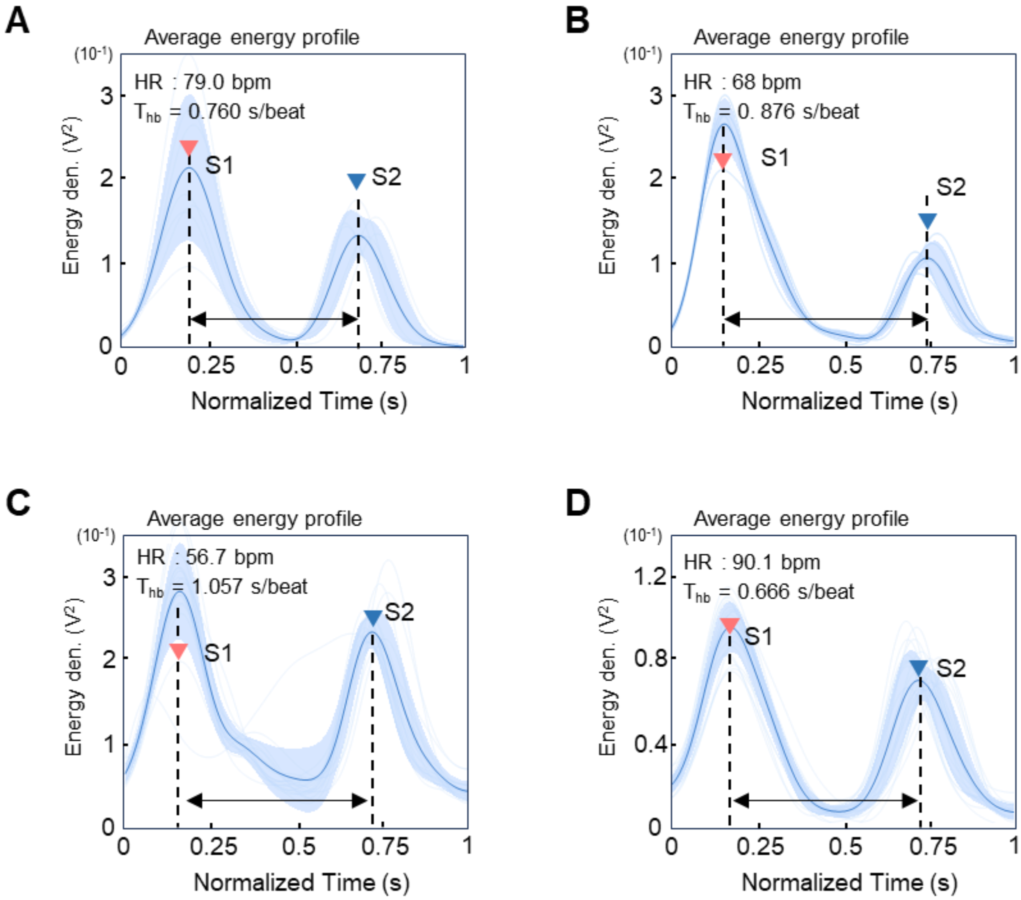


Fig. S20. Cardiac-pattern extraction across healthy participants.

Table S1. Clinical study participant demographics, recruitment criteria, and study objectives.

	Healthy participants	**Clinical participants**	
		Adults	Pediatric
Age (years) (Mean ± SD)	21 – 32 (27.6 ± 4.7)	26 – 86 (72 ± 8)	8 – 16 (10.7 ± 3.6)
Number of participants	5	32	8
Gender			
Male	4	26	5
Female	1	6	1
Inclusion Criteria	• Age ≥ 18 years • Signed a written informed consent • Not taking any prescription medications • No known medical diagnosis or chronic conditions	• Age 19 – 90 years • Signed a written informed consent • Diagnosed respiratory disease • Patients receiving outpatient or inpatient care	• Age 6-17 years • Diagnosis of persistent asthma • Presenting or being treated for an acute asthma exacerbation or wheezing • Parent is able to speak and understand English
Study objective	• Diagnosis model development	• Diagnosis model development	• Diagnosis model development • Longitudinal progress monitoring
Sound type	• Healthy	• Wheezing • Rhonchi • Crackle	• Wheezing

Table S2. Clinical information of the 38 patients used for model training.

#	Age	Gender	Measurement Location	Sound type	Diagnosis
1	86	M	P, LUL	Wheezing, Rhonchi	Asthma
2	82	M	P, LUL	Wheezing	COPD, Asthma
3	68	F	P, LUL	Wheezing	Bronchial asthma, COPD
4	81	M	P, RLL	Wheezing	Lung cancer COPD
5	69	M	P, RLL	Crackle	Lung cancer
6	78	M	P, RLL	Crackle	Pneumonia
7	57	F	P, RLL	Crackle	Pneumonia COPD
8	74	M	P, RLL	Rhonchi	Pneumonia
9	68	M	P, RLL	Crackle	CPFE
10	59	F	P, LLL	Rhonchi	Bronchiectasis, Asthma, COPD
11	76	F	P, RLL	Crackle	IPF, Pneumonia, Lung cancer
12	68	M	P, RLL	Crackle	IPF
13	71	M	P,LLL	Wheezing	COPD
14	68	M	P,RLL	Crackle	CPFE
15	63	M	P,RLL	Crackle	CPFE
16	69	M	P,RLL	Crackle	CPFE
17	80	M	P,RLL	Wheezing	Asthma
18	75	M	P,RLL	Crackle	IPF
19	68	M	P,RLL	Crackle	IPF
20	70	M	A, LUL	Wheezing	Bronchial asthma
21	72	F	P, RLL	Wheezing	Lung cancer COPD

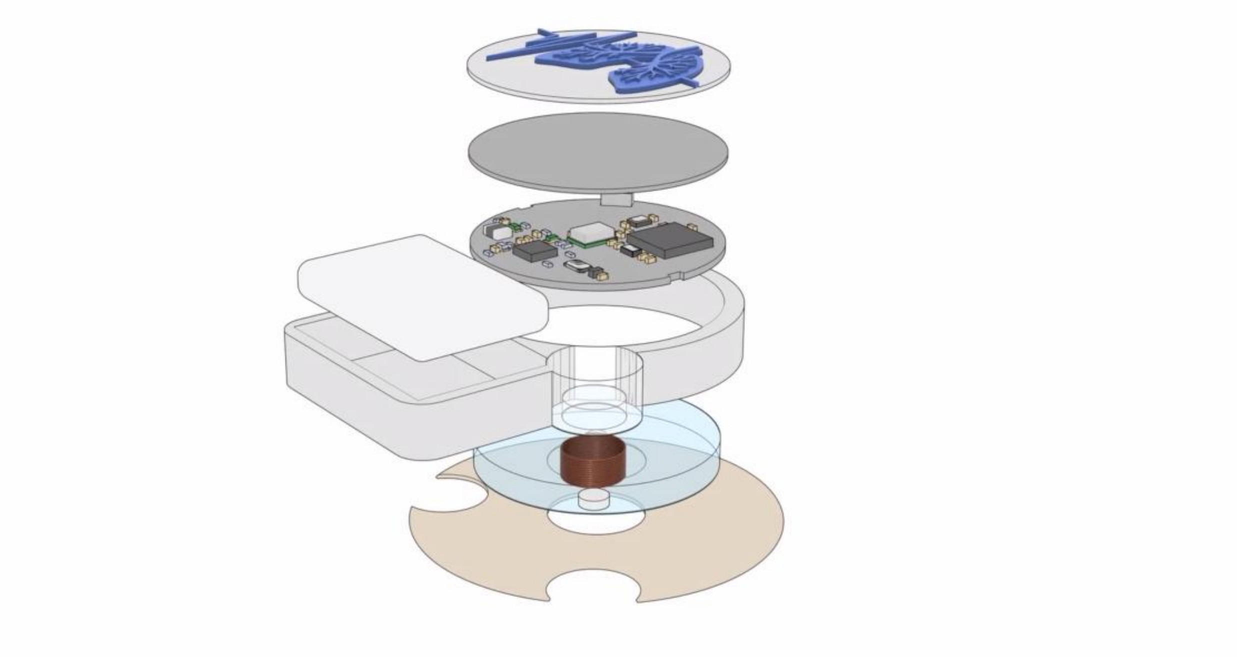
22	26	M	P, LLL	Wheezing	Pneumonia Asthma
23	72	M	P,RLL	Crackle	Pneumonia
24	83	M	P, RLL	Crackle	IPF
25	71	M	P, RLL	Crackle	NSIP
26	80	M	P, RUL	Wheezing	Lung cancer
27	63	M	P, LLL	Wheezing	COPD
28	75	M	P, RLL	Wheezing	COPD
29	80	M	P, LLL	Wheezing	COPD
30	78	F	P, RLL	Crackle	Pneumonia
31	75	M	P, LUL	Crackle	Pneumonia
32	80	M	P, LUL	Wheezing	COPD
33	8	M	A, RUL	Wheezing	Asthma
34	15	M	A, RUL	Wheezing	Asthma
35	9	M	A, RUL	Wheezing	Asthma
36	16	F	A, RUL	Wheezing	No pulmonary diagnosis
37	8	M	A, RUL	Wheezing	No pulmonary diagnosis
38	8	M	A, RUL	Wheezing	Asthma

IPF (Idiopathic Pulmonary Fibrosis), CPFE (Combined pulmonary Fibrosis and Emphysema)
NSIP (Nonspecific Interstitial Pneumonia), P (Posterior), A (Anterior)
LLL/RLL (Left/Right Lower Lobe), LUL/RUL (Left/Right Upper Lobe)

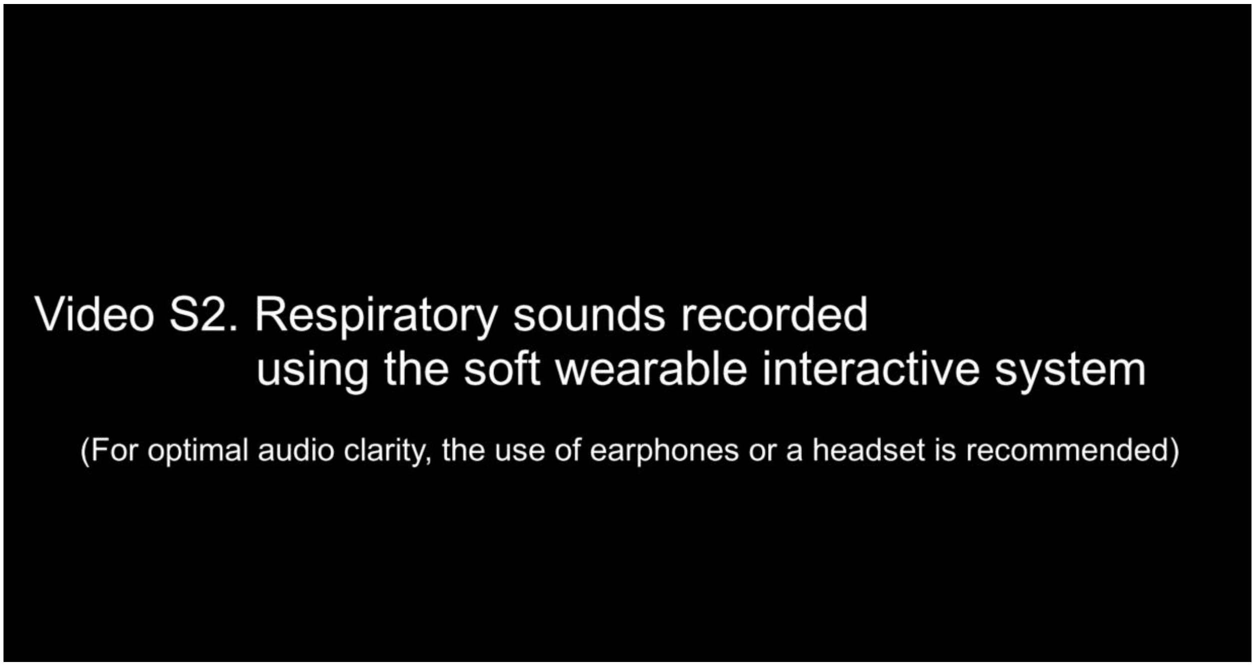
Table S3. Comparative summary research on respiratory sensing systems.

Reference	Continuous Monitoring	Interactive feedback	SNR	Measurement (sensor)	Physiological Data	Clinical Study (# of patients)	ML model
This work	Yes	Yes	30 dB	Sound	Hearth Rate Respiratory Rate Breathing Patterns (Healthy, Wheezing, Crackles, Rhonchi)	Patients with lung disease (38)	Multiclass Classification (>85%) Real-time model deployment
[17]	Yes	No	26.23 dB	Sound (Mic)	Breathing Patterns Classifications	Patients with lung disease (20)	CNN (84.3 %)
[43]	Yes	No	26 dB (noise 90 dB)	Sound (Mic) Motion / Posture (IMU)	Heart Rate Respiratory Rate Breathing Intensity Breathing Intervals Posture	Cardiorespiratory sound analysis Bowel sound analysis (70)	-
[44]	Yes	No	21 dB	Sound (Mic)	Breathing Patterns Classifications	Pediatric asthma patients (2) Elderly patients with COPD (5)	CNN (>80 %)
[45]	No	No	-	Sound (4 multi-channel)	Breathing Patterns Classifications	Asthma (30) Normal (30)	SVM classifiers
[35]	No	No	36 dB	Sound (Triboelectric)	Breathing Patterns Classifications	No clinical study Normal (9)	CNN (> 97 %)
[46]	No (Single RR per night)	No	-	PPG	Respiratory Rate (Algorithm)	Sleep Apnea (28)	-

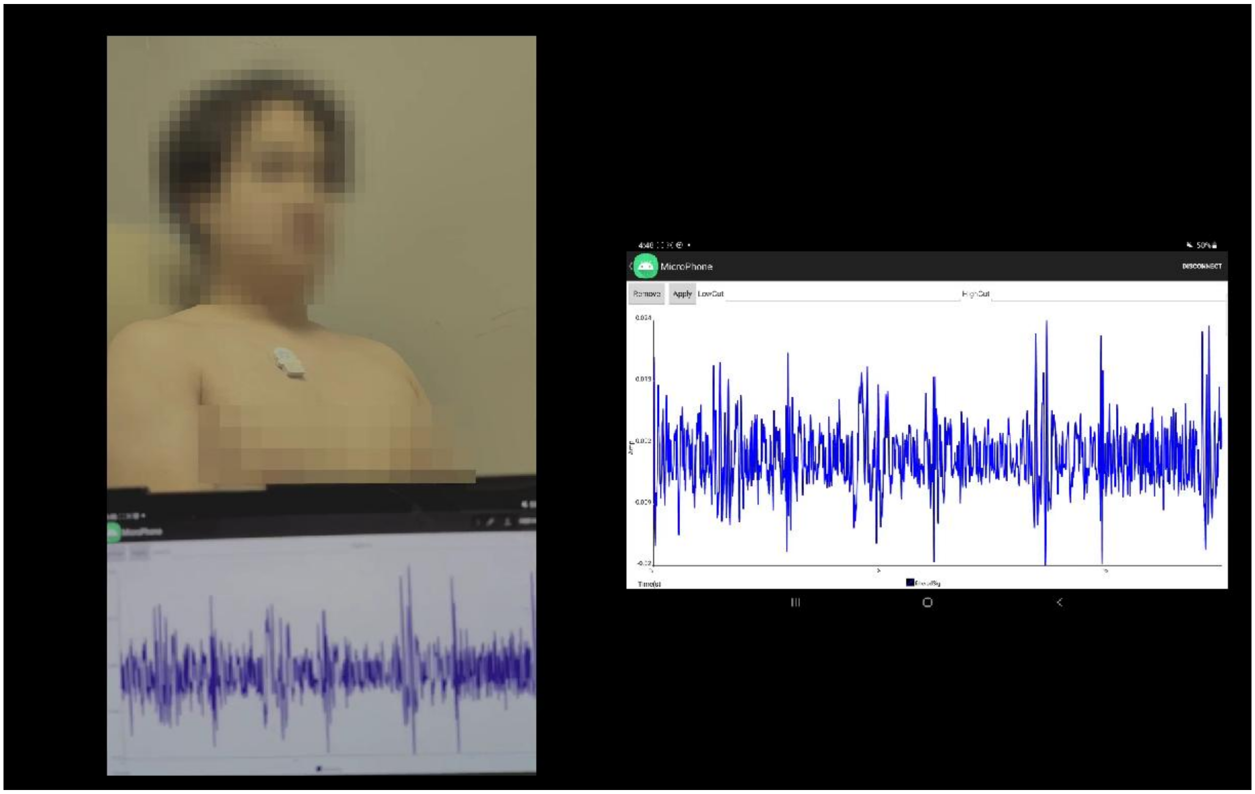
[47]	Yes	No	-	RF sensing	Hear Rate Viability	6222	-
[48]	No		-	ECG, IMU	Respiratory Rate (Algorithm)	Normal (15)	SVM Gaussian



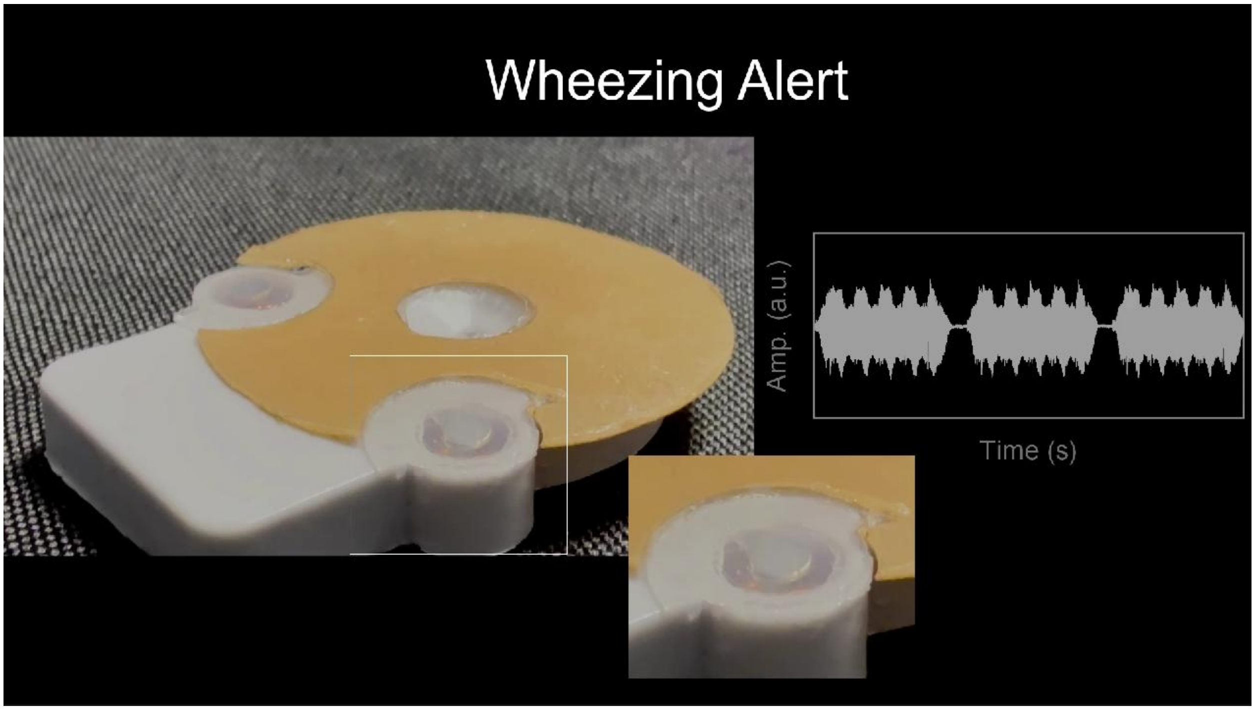
Movie S1. Overview of the soft wearable interactive system.



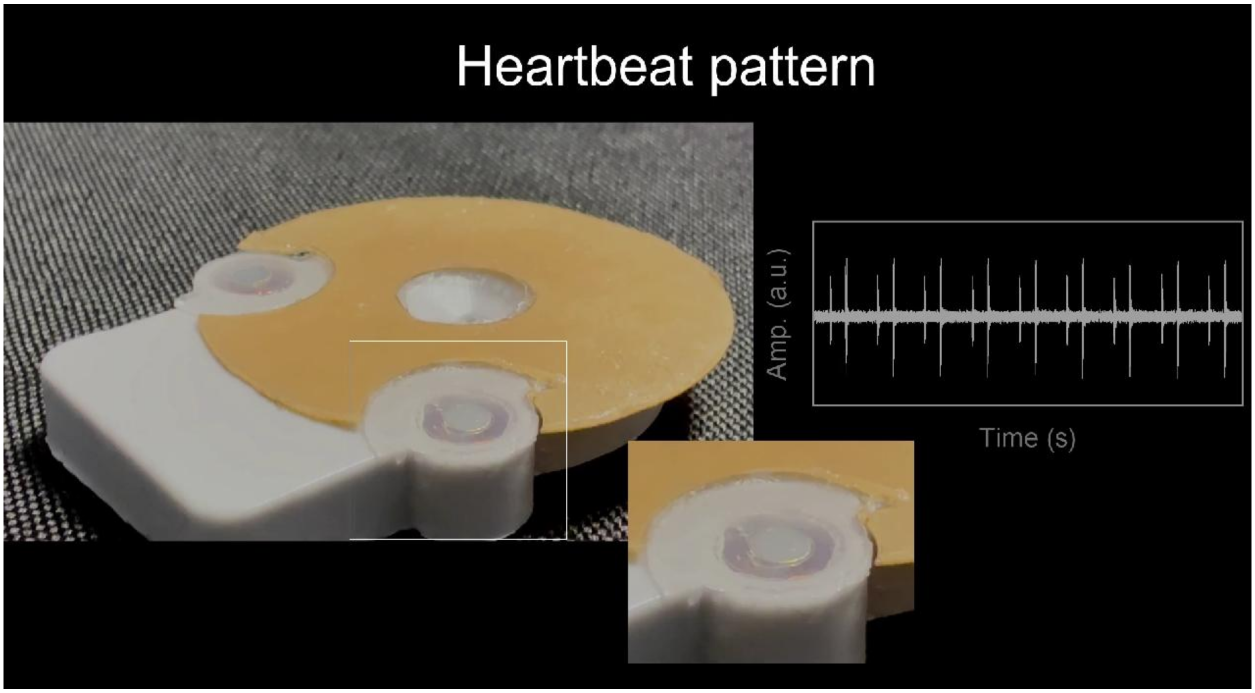
Movie S2. Respiratory sounds recorded using the soft wearable interactive system.



Movie S3. Real-time auscultation data acquisition using a custom-designed user interface.



Movie S4. Distinct vibrotactile feedback patterns generated based on respiratory diagnosis.



Movie S5. Vibrotactile reproduction of cardiac patterns.